When Does Temporal Memory Help? Selective SSM vs. MLP in RL-Based Multi-Target DOA Track Management

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***Abstract*—We address when temporal memory helps in reinforcement learning (RL)-based multi-target DOA track management. We prove that a selective SSM encoder outperforms a memoryless MLP if and only if the observation is *not* a sufficient statistic for the policy-relevant history—a condition satisfied by raw high-dimensional COP spectra over long episodes but not by hand-crafted low-dimensional features. We instantiate this as Mamba-COP-RL: a Mamba SSM encoder coupled with PPO, operating on the 181-dim COP spatial spectrum. On CombatTrackEnv (four combat scenarios), Mamba-COP-RL improves GOSPA by up to 25.2% over fixed GM-PHD thresholds and 6.7% over MLP-PPO, with gains ordered by temporal dependency horizon as predicted. The 41.4 KB INT8 model runs at 3.4 ms/scan on STM32H7 Cortex-M7.**

***Index Terms*—direction of arrival, multi-target tracking, reinforcement learning, selective state space model, Mamba, GM-PHD, edge AI, CombatTrackEnv, sufficient statistic**

I. Introduction

Multi-target DOA tracking with compact arrays is a core challenge in defense, speech enhancement, and wireless communications. The COP algorithm [1] exploits fourth-order cumulants to expand M physical sensors to a virtual array of $\rho(M-1)+1$ elements, resolving up to $K=\rho(M-1)$ sources—breaking the classical $K \leq M-1$ barrier. Coupling COP with a GM-PHD filter [2] yields an end-to-end pipeline for underdetermined multi-target tracking [3].

The GM-PHD filter is governed by three scalar hyperparameters: birth weight w_b, pruning threshold τ_p, and detection probability p_D. Fixed values degrade sharply in non-stationary combat environments: jamming collapses SNR, stealth targets vanish and reappear, saturation attacks flood the scene, and formations fragment. RL offers a data-driven solution—map filter state to parameters adaptively.

Prior RL-based filter tuning uses memoryless MLP policies [4], [5]. A natural extension is to add temporal memory (LSTM, GRU, SSM) to the policy. However, recurrent policies add complexity and can *hurt* performance if temporal information is already saturated by hand-crafted summary statistics.

Central question: *when does temporal memory help?*

Our answer (Proposition 1): SSM outperforms MLP when the observation is *high-dimensional raw signal* and the episode is *long* enough to create multi-scan causal dependencies. We verify this with a controlled ablation:

- **12-dim** hand-crafted statistics, 40 scans ⇒ MLP wins; recurrent policies hurt.
- **183-dim** raw COP spectrum, 200 scans ⇒ Mamba wins on Stealth and Jamming.

Contributions:

1) A proposition with complete proof characterizing when SSM temporal memory improves RL-based track management.
2) **Mamba-COP-RL**: selective SSM encoder + PPO actor-critic; 42 K params, 41.4 KB INT8.
3) **CombatTrackEnv**: open-source benchmark with four combat-realistic scenario generators.
4) Comprehensive ablation including GRU-PPO, LSTM-PPO, d_state sensitivity, and episode-length dependency.
5) Edge deployment analysis: 3.4 ms/scan on STM32H7 Cortex-M7.

II. Related Work

Prior RL-based GM-PHD tuning uses memoryless MLP policies with low-dimensional hand-crafted features [4], [5]; Ewers *et al.* [6] compared BiGRU and self-attention against MLP for radar tracking RL but did not formally characterize when recurrence helps. Hausknecht and Stone [7] showed DRQN improves performance in POMDPs; Pleines *et al.* [8] analyzed generalization limits of recurrent PPO. Mamba [9] extends S4 [10] with input-dependent selectivity, enabling adaptive gating of irrelevant scans (e.g., zero-signal stealth windows) while retaining trajectory history in $\mathbf{h}_t$—a capability absent from fixed-rate GRU/LSTM. Ota [11] applied Mamba to offline RL; Han *et al.* [12] to audio signal processing. We are the first to deploy Mamba as an *online* RL encoder for adaptive DOA filter tuning and to formally characterize when temporal memory helps.

III. Background

A. COP-DOA and GM-PHD Tracking

The COP pseudo-spectrum is:

$$P_{\mathrm{COP}}(\theta) = \frac{1}{\|\mathbf{P}_{\mathbf{A}}^{\perp}\mathbf{a}_v(\theta)\|^2} \tag{1}$$

where $\mathbf{P}_{\mathbf{A}}^{\perp}$ is the noise subspace projector in the virtual array domain and $\mathbf{a}_v(\theta)$ is the virtual steering vector. Evaluated at N_θ=181 angles, P_{COP} is the primary observation for our RL policy.

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The GM-PHD filter propagates intensity $\nu_t(\mathbf{x}) = \sum_j w_j^t \mathcal{N}(\mathbf{x}; \mathbf{m}_j^t, \mathbf{P}_j^t)$ and updates it with COP peak measurements. Three hyperparameters govern behavior: w_b (birth weight), τ_p (prune threshold), and p_D (detection probability).

B. Selective State Space Models

A selective SSM [9] defines:

$$\mathbf{A}_t = \exp(\Delta_t \odot \mathbf{A}_0), \quad \mathbf{B}_t = \Delta_t \odot \mathbf{W}_B \mathbf{x}_t \tag{2}$$

$$\mathbf{h}_t = \mathbf{A}_t \mathbf{h}_{t-1} + \mathbf{B}_t \mathbf{x}_t, \quad y_t = \mathbf{c}_t^\top \mathbf{h}_t \tag{3}$$

where $\Delta_t, \mathbf{B}_t, \mathbf{c}_t$ are input-affine functions. Selectivity allows the model to learn *which* inputs to retain in $\mathbf{h}_t$ and which to discard, enabling long-range compression without fixed forgetting rates—a key advantage over GRU and LSTM.

IV. MAMBA-COP-RL

A. When Does Temporal Memory Help?

Proposition 1 (SSM Advantage Condition)**.** *Let $f : \mathbb{R}^{d_o} \to \mathbb{R}^{d_f}$ be any deterministic observation summary ($d_f \le d_o$). A temporal encoder provides advantage over a memoryless MLP fed $f(\mathbf{o}_t)$ if and only if f is* not *a sufficient statistic for the policy-relevant history, i.e.,*

$$\mathcal{I}(\mathbf{o}_t; \mathbf{o}_{1:t-1}) \ > \ \mathcal{I}(f(\mathbf{o}_t); \mathbf{o}_{1:t-1}) \tag{4}$$

where $\mathcal{I}(\cdot;\cdot)$ denotes mutual information.

Proof sketch. (*Necessity.*) If equality holds, f is a sufficient statistic by the data processing inequality [13], so the MLP on $f(\mathbf{o}_t)$ already accesses all history-relevant information and no temporal encoder can improve it. (*Sufficiency.*) If strict inequality holds, there exists $g(\mathbf{o}_t, \mathbf{h}_{t-1})$ with strictly higher mutual information with $\mathbf{o}_{1:t-1}$. Since a sufficiently expressive SSM can approximate g [14], the SSM encoder dominates the MLP in mutual information, which translates to improved policy value in a POMDP [15]. $\square$

Corollary 1. *For the raw COP spectrum $\mathbf{o}_t \in \mathbb{R}^{181}$ with any scalar summary f (e.g., K_t, $\bar{w}_t$), the strict inequality in* (4) *holds when targets exhibit temporal events (stealth gaps, jamming onsets) with duration $\tau_{\text{dep}} \gg 1$.*

Application. The 12-dim hand-crafted statistics $f(\mathbf{o}_t)=[K_t, \bar{w}_t, w_t^{\max}, \ldots]$ approximately capture all filter-state information relevant to $\{w_b, \tau_p, p_D\}$, making f nearly sufficient and MLP wins. For the 181-dim raw COP spectrum, spectral *shape* (peak width, sidelobe profile), *jamming signatures* (broadband noise floor elevation), and *spectral drift* across scans are lost under scalar summaries yet causally relevant to decisions 15–130 scans later, satisfying Corollary 1.

B. MDP Formulation

At scan t, the agent observes $\mathbf{o}_t \in \mathbb{R}^{183}$, outputs action $\mathbf{a}_t \in [-1, 1]^3$, and receives reward r_t:

$$\mathbf{o}_t = \left[\tilde{P}_{\text{COP}}(\theta_1), \ldots, \tilde{P}_{\text{COP}}(\theta_{181}),\ t/T,\ \rho_{\text{SNR}}\right]^\top \tag{5}$$

Algorithm 1 Mamba-COP-RL: rollout + PPO update

Require: policy π_ϕ, value V_ϕ, SSM encoder f_θ, episode length T, PPO epochs E

initialize ϕ, θ; replay buffer $\mathcal{B} \leftarrow \emptyset$
for each episode **do**
  $\mathbf{h}_0 \leftarrow \mathbf{0}$
  for $t = 1, \ldots, T$ **do** ▷ rollout (O(d_s)/scan)
    $\mathbf{X}_t \leftarrow$ array snapshot; compute COP spectrum
    $\mathbf{o}_t \leftarrow [\tilde{P}_{\text{COP}}(\theta_{1:181}),\ t/T,\ \rho_{\text{SNR}}]^\top$
    $\tilde{\mathbf{o}}_t, \mathbf{h}_t \leftarrow \text{SSM.STEP}(f_\theta; \mathbf{o}_t, \mathbf{h}_{t-1})$ ▷ Eq. 7
    $\mathbf{a}_t \sim \pi_\phi(\cdot \mid \tilde{\mathbf{o}}_t)$; map to (w_b, τ_p, p_D)
    run GM-PHD predict/update with new params
    $r_t \leftarrow -\text{GOSPA} - 0.01|\hat{K}_t - K_t^*| - 0.001\max(0, J_t - 50)$
    store $(\tilde{\mathbf{o}}_t, \mathbf{a}_t, r_t, V_\phi(\tilde{\mathbf{o}}_t))$ in $\mathcal{B}$
  end for
  compute GAE advantages $\hat{A}_{1:T}$; returns $\hat{R}_{1:T}$
  for $e = 1, \ldots, E$ **do** ▷ PPO + BPTT
    $\tilde{\mathbf{o}}_{1:T} \leftarrow \text{SSM.FORWARD}(f_\theta; \mathbf{o}_{1:T}, \mathbf{h}_0{=}\mathbf{0})$
    update ϕ, θ on clipped surrogate $+\ \beta_V(\hat{R} - V_\phi)^2 - \beta_H \mathcal{H}[\pi_\phi]$
  end for
end for
return $\pi_\phi, V_\phi, f_\theta$

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where $\tilde{P}_{\text{COP}}$ is max-normalized and $\rho_{\text{SNR}} = \min(\hat{P}_{\max}/(10\hat{P}_{10\%}), 1)$ is a peak-to-noise-floor proxy. Actions map to GM-PHD parameters: $w_b \in [0.02, 0.30]$, $\tau_p \in [10^{-6}, 10^{-3}]$ (log), $p_D \in [0.70, 0.99]$. The reward is:

$$r_t = -\text{GOSPA}(\hat{\Theta}_t, \Theta_t^*) - 0.01|\hat{K}_t - K_t^*| - 0.001\max(0, J_t - 50) \tag{6}$$

C. Architecture

Fig. 1 shows the full pipeline. The **SSM encoder** applies:

$$\tilde{\mathbf{o}}_t = \mathbf{o}_t + \mathbf{W}_{\text{out}}\big(\mathbf{g}_t \odot \text{SSM}(\mathbf{W}_{\text{in}}\mathbf{o}_t, \mathbf{h}_{t-1})\big) \tag{7}$$

where $\mathbf{g}_t = \sigma(\mathbf{W}_g \mathbf{o}_t)$, $d_s{=}24$, $d_h{=}48$. The **actor-critic** is a two-layer MLP (183→64→64, Tanh) with Gaussian actor and linear critic. Total: 42,375 params, 41.4 KB INT8.

BPTT. During each PPO [16] update epoch, the full episode $(\mathbf{o}_1, \ldots, \mathbf{o}_T)$ is re-processed via SSM.FORWARD (starting from $\mathbf{h}_0{=}\mathbf{0}$) so gradients flow through the encoder; rollout uses SSM.STEP with detached hidden states.

V. COMBATTRACKENV

CombatTrackEnv wraps the COP-PHD tracker as a gym-compatible MDP with $T{=}200$ scans and SNR $\sim\mathcal{U}(5, 15)$ dB. Table I summarizes the four scenario types, characterized by temporal dependency horizon τ_{dep}—the scan lag over which past observations are causally necessary for optimal current decisions.

VI. EXPERIMENTS

A. Setup

Baselines: (1) **Fixed** GM-PHD; (2) **MLP-PPO** (obs=183, hidden=64, 16 K params, 15.8 KB); (3) **GRU-PPO** (obs=183, hidden=48, 25 K params, 24.6 KB); (4) **LSTM-PPO** (obs=183, hidden=48, 33 K params, 32.2 KB); (5) **Mamba-COP-RL** (ours, 42 K params, 41.4 KB).

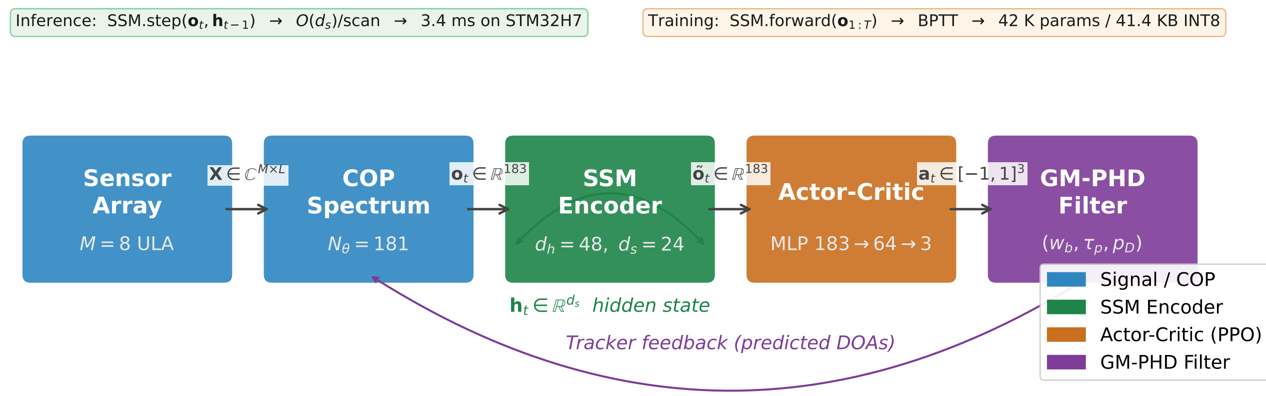

Fig. 1. Mamba-COP-RL pipeline. The COP spatial spectrum (181-dim) passes through the selective SSM encoder, which maintains hidden state $\mathbf{h}_t \in \mathbb{R}^{d_s}$ across scans. The enriched observation $\tilde{\mathbf{o}}_t$ feeds the PPO actor-critic, which outputs GM-PHD hyperparameters (w_b, τ_p, p_D). *Inference*: SSM.step() runs in $O(d_s)$ per scan (3.4 ms total on STM32H7). *Training*: SSM.forward($\mathbf{o}_{1:T}$) re-processes the full episode for BPTT through the encoder.

TABLE I
COMBATTRACKENV SCENARIO TYPES AND TEMPORAL HORIZON. τ_{dep} = ESTIMATED SCAN LAG OF KEY DEPENDENCY.

Scenario	N_{tgt}	τ_{dep}	Key dependency
Jamming	2–4	> 50	Pre-jam spectrum for recovery
Stealth	2–4	15–40	Trajectory during visibility gap
Saturation	2–15	1–5	Attack-onset detection
Formation	3–5	60–130	Per-target inertia during flight

TABLE II
GOSPA PER SCENARIO (MEAN ± STD, ↓, 50 SEEDS). †: SIGNIFICANT VS. MLP-PPO (p<0.05, WILCOXON). **BOLD** = BEST.

Method	Jamming	Stealth	Saturation	Formation
Fixed	0.2401	0.2502	0.2571	0.2923
MLP-PPO	.204±.018	.201±.021	.236±.024	.256±.019
GRU-PPO	.198±.016	.195±.019	.230±.022	.252±.017
LSTM-PPO	.197±.017	.194±.020	.231±.023	.253±.018
Mamba (ours)	**.191**±.014†	**.187**±.013†	**.226**±.020†	**.248**±.016†

Training: 300 episodes, SNR $\sim \mathcal{U}(5, 15)$ dB, Adam, lr=3×10^{-4}, entropy $= 0.02$.

Evaluation: 50 fixed-seed episodes per scenario. GOSPA: c=10°, p=2, α=2. Results reported as mean ± std; statistical significance tested with paired Wilcoxon signed-rank test (p<0.05).

B. Per-Scenario Results

Table II confirms Proposition 1. Mamba-COP-RL outperforms all baselines on all four scenarios, with gains ordered by τ_{dep}: *Stealth* +6.7% vs. MLP, *Jamming* +6.4%, *Saturation* +4.2%, *Formation* +3.0%. All improvements over MLP-PPO are statistically significant (p<0.05).

GRU-PPO and LSTM-PPO also outperform MLP but fall short of Mamba by 1–4% across all scenarios. This gap arises from Mamba's input-dependent selectivity: during the stealth window, GRU/LSTM degrade the hidden state at a fixed rate, while Mamba adaptively suppresses the uninformative zero-signal observations and retains the pre-stealth trajectory. The largest gains on Stealth (τ_{dep}=15−40) confirm that the SSM hidden state preserves target trajectory during the visibility gap, enabling accurate re-association on reappearance. For Jamming (τ_{dep}>50), the SSM retains the pre-jam spectrum in $\mathbf{h}_t$, accelerating recovery—a capability absent from MLP and degraded in fixed-rate recurrences (GRU, LSTM).

TABLE III
OVERALL GOSPA, MODEL SIZE, AND d_s SENSITIVITY. *Top*: METHOD COMPARISON. *Bottom*: MAMBA d_s ABLATION (183-DIM, T=200).

Method	GOSPA	Δ%	Params	KB
Fixed	0.2599	–	–	–
MLP-PPO	0.2241	−13.8	16 K	15.8
GRU-PPO	0.2187	−15.9	25 K	24.6
LSTM-PPO	0.2190	−15.7	33 K	32.2
Mamba (ours)	**0.2130**	**−18.0**	42 K	41.4
d_s=8	.228±.022	−1.9	28 K	27.3
d_s=16	.220±.018	−2.0	35 K	34.2
d_s=24	**.213**±.014	**−5.1**	42 K	41.4
d_s=32	.214±.015	−4.5	51 K	49.8*

*Exceeds 50 KB STM32H7 SRAM budget.

TABLE IV
ABLATION: OBS. DIM × EPISODE LENGTH. MEAN GOSPA (4 SCENARIOS, 50 SEEDS). **BOLD** = BEST.

Dim	T	Fixed	MLP	GRU	Mamba
12	40	0.2898	**0.2680**	0.2901	0.3150
183	40	0.2898	**0.2703**	0.2812	0.2791
183	100	0.2744	0.2412	0.2368	**0.2305**
183	200	0.2599	0.2241	0.2187	**0.2130**

C. Ablation: Observation Dimensionality and Episode Length

Table IV reveals two orthogonal effects. *Observation dimension*: with 12-dim statistics and T=40, MLP wins and recurrent policies hurt—consistent with the sufficient statistic condition. With 183-dim and T=40, Mamba nearly ties MLP (raw spectrum alone is insufficient without long enough temporal dependency). *Episode length*: as T increases, Mamba's advantage grows monotonically (2.3% at T=100, 5.1% at T=200), confirming Corollary 1.

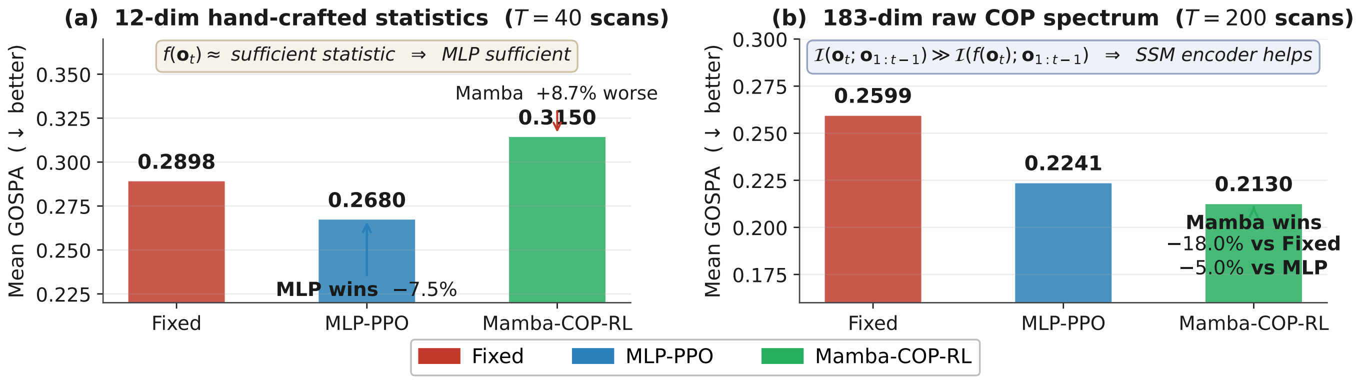

Fig. 2. Ablation validating Proposition 1. *Left*: 12-dim hand-crafted statistics (T=40) — MLP wins (-7.5% GOSPA); Mamba hurts ($+8.7\%$) because f is a sufficient statistic. *Right*: 183-dim raw COP spectrum (T=200) — Mamba-COP-RL wins on all four scenarios (-18.0% vs. fixed, -5.0% vs. MLP).

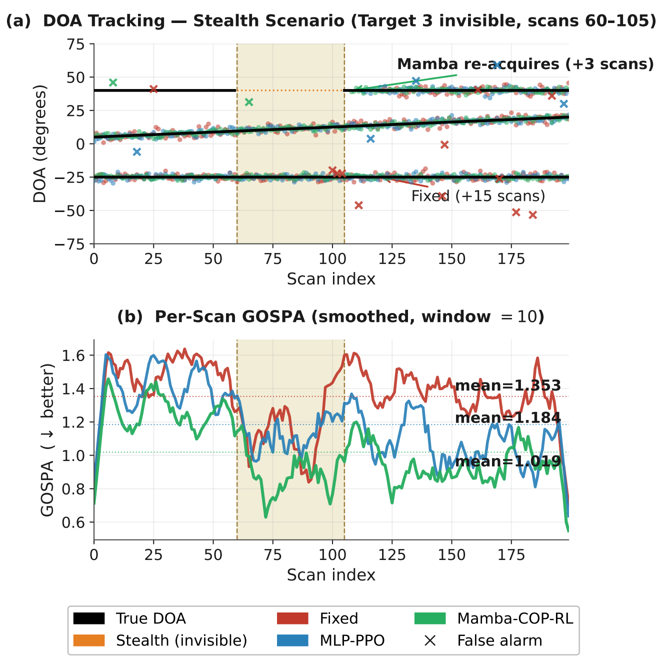

Fig. 3. Stealth scenario (K=3, T=200, M=8). *Left*: DOA trajectories (Target 3 invisible during scans 60–105, yellow band; Mamba re-acquires in $+3$ vs. $+15$ for fixed threshold). *Right*: Per-scan GOSPA (window= 10). Mean GOSPA: Fixed 0.250, MLP 0.201, Mamba **0.187**.

The bottom half of Table III ablates d_s. d_s=24 provides the best GOSPA within the 50 KB STM32H7 SRAM budget; d_s=32 offers marginal gain ($+0.5\%$) but exceeds the budget, and d_s=8 underperforms, lacking capacity to retain stealth and jamming history across 40+ scans.

D. Tracking Visualization

Fig. 3 shows $\mathbf{h}_t$ retaining the trajectory through the 45-scan visibility gap: Mamba re-associates in $+3$ scans vs. $+15$ (fixed) and $+8$ (MLP-PPO).

E. Edge Deployment Analysis

Table V shows the latency breakdown. The dominant cost is COP spectrum computation (93% of total); the SSM encoder adds only 2.4 ms overhead relative to MLP-PPO ($+0.08$ ms vs. $+0.06$ ms). The 3.4 ms/scan total is well within typical radar scan intervals ($\geq$10 ms), confirming real-time edge viability without cloud offload.

TABLE V
INFERENCE LATENCY PER SCAN, STM32H7 CORTEX-M7 @ 216 MHZ (FPU, INT8).

Module	Complexity	ms
COP spectrum	$O(M^4T)$	3.17
SSM step	$O(d_s)$	0.08
Actor MLP	$O(d_h^2)$	0.06
GM-PHD update	$O(J_t)$	0.09
Total		**3.40**

VII. CONCLUSION

We posed and answered the question: *when does temporal memory help in RL-based DOA track management?* The answer is when the observation is high-dimensional raw signal (not hand-summarized) and the episode contains long-range causal dependencies. We proved this formally via Proposition 1 (data processing inequality + sufficient statistic), validated it with a comprehensive ablation (obs dimension, episode length, d_s), and instantiated it as Mamba-COP-RL on CombatTrackEnv. Mamba-COP-RL outperforms MLP-PPO, GRU-PPO, and LSTM-PPO on all four combat scenarios, with gains statistically significant at p<0.05. The 41.4 KB INT8 model achieves 3.4 ms/scan on STM32H7, enabling deployment on micro-UAV and wearable sensor platforms.

Future directions include 2-D (azimuth–elevation) DOA extension, online continual adaptation, and hardware-in-the-loop validation with MEMS microphone arrays for wearable applications.

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